

GROUP 6

Fraud Detection - Machine Learning

Track	Machine Learning
Duration	2 Weeks (10 working days, 4 hours/day)
ML Task	Classification
Classification Type	Binary Classification
Target	isFraud: 0 = Normal, 1 = Fraud
User Interface	Required

Project Analysis

Financial transaction systems process very large numbers of transactions, making manual review difficult. This project requires students to build a machine-learning model that learns from labeled transaction data and predicts whether a transaction is potentially fraudulent or normal.

The focus is the complete ML workflow: data understanding, class imbalance, preprocessing, model comparison, evaluation, and UI integration.

What the Student Must Build (Mandatory)

- Load and inspect the fraud dataset.
- Understand the target isFraud: 0 = Normal and 1 = Fraud.
- Check missing values, duplicates, data types, and class balance.
- Perform EDA comparing fraud and normal transactions.
- Prepare features and encode categorical variables where required.
- Identify and avoid data leakage.
- Split data into training and testing sets using an appropriate strategy.
- Train at least two classification models.
- Evaluate with Precision, Recall, F1-score, Confusion Matrix, and Accuracy.
- Do not select a model using Accuracy alone because fraud datasets can be highly imbalanced.
- Compare models and justify the final selection.
- Save the final preprocessing/model pipeline.
- Build a functional UI for entering transaction information and displaying Normal / Potential Fraud.
- Display probability/confidence if supported.

Technical Requirements

- Python
- Pandas
- NumPy
- Scikit-learn
- Matplotlib or equivalent

- Joblib/Pickle
- Git & GitHub
- Streamlit/Gradio or another suitable UI framework

Dataset Requirements

- Recommended dataset: PaySim - Synthetic Financial Datasets for Fraud Detection.
- Target: isFraud.
- Classes: 0 = Normal transaction; 1 = Fraudulent transaction.
- Typical fields: step, type, amount, nameOrig, oldbalanceOrg, newbalanceOrig, nameDest, oldbalanceDest, newbalanceDest, isFraud, isFlaggedFraud.
- Because the dataset is large, the instructor may provide a manageable prepared subset while preserving enough fraud cases for training and evaluation.
- Students must review balance/flag fields for possible leakage before training.

Optional Features (Bonus)

These features may improve the project but are not required for basic project completion.

- Feature importance visualization
- SHAP/explainability
- Batch CSV prediction
- Advanced imbalance handling
- Fraud analytics dashboard
- PDF report

References and Starting Resources

- Kaggle - PaySim: Synthetic Financial Datasets for Fraud Detection
- Scikit-learn Documentation
- Pandas Documentation